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Tutorial 8 - Simulating Atherosclerotic Plaque

In this tutorial, we were tasked with using our computational fluid dynamics (CFD) skills to investigate the impact of atherosclerotic plaque on blood flow through an artery. This assignment aimed to explore how different plaque geometries alter velocity fields and wall shear stress, and to consider the physiological consequences of such changes. To simulate this scenario, I created two simplified artery models in SolidWorks. The first was a baseline model, a straight cylindrical tube representing a healthy artery segment. The second model included a plaque-shaped occlusion, which I modeled with a notch taken out in the Solidworks modeling to represent partial blockage as a plaque would once the model was run into Ansys Fluid Flow (Fluent). This shape was chosen to simulate a smooth but progressive buildup, allowing for controlled narrowing while avoiding sharp geometric transitions that might introduce numerical artifacts.

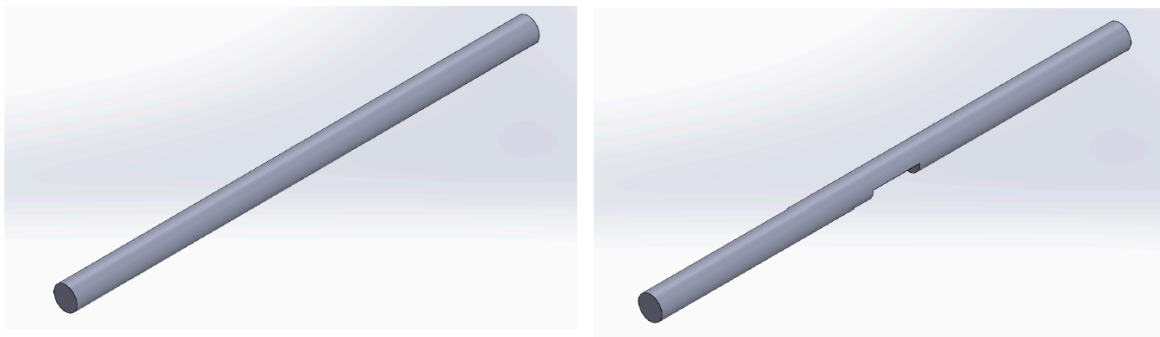


Figure 1: Normal Healthy Artery (Baseline) vs Artery With an Occlusion Due to Plaque Buildup

Both geometries were imported into ANSYS Fluent for steady-state CFD analysis. I used a structured mesh with refinement near the walls and around the occlusion to capture the steep velocity gradients expected in those regions. Blood was modeled as an incompressible, Newtonian fluid with a density of 1060 kg/m^3 and a viscosity of $0.0035 \text{ Pa}\cdot\text{s}$, consistent with prior class examples. Boundary conditions included a uniform inlet velocity of 0.2 m/s and a zero-pressure outlet, while walls were set to no-slip conditions.

The healthy artery's flow remained laminar and fully developed with a classic parabolic profile. This confirmed that the simulation setup and boundary conditions were functioning as intended. In the plaque model, the velocity profile was dramatically altered. As expected, the narrowing caused an increase in velocity through the stenosed region. This acceleration is consistent with conservation of mass and mimics the hemodynamic effects seen in clinical atherosclerosis.

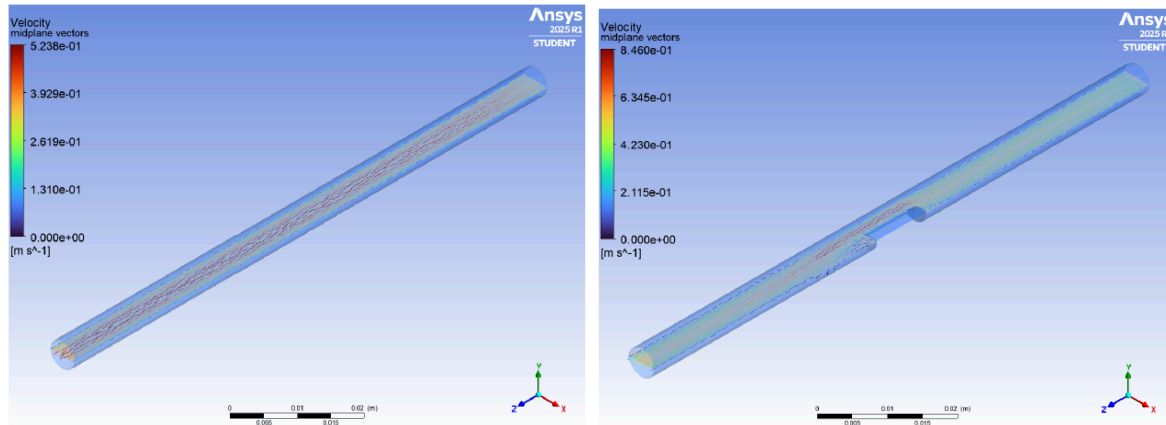


Figure 2: Velocity Fields of Normal Healthy Artery vs Artery With an Occlusion Due to Plaque Buildup

Wall shear stress in the healthy model remained relatively uniform, peaking at the entrance before settling into a steady-state profile along the walls. In contrast, the plaque model showed a marked spike in WSS at the narrowest point of the plaque. This elevated shear stress is concerning from a physiological perspective: high WSS is known to damage endothelial cells and trigger biological responses that can destabilize plaques, potentially leading to rupture or thrombosis.

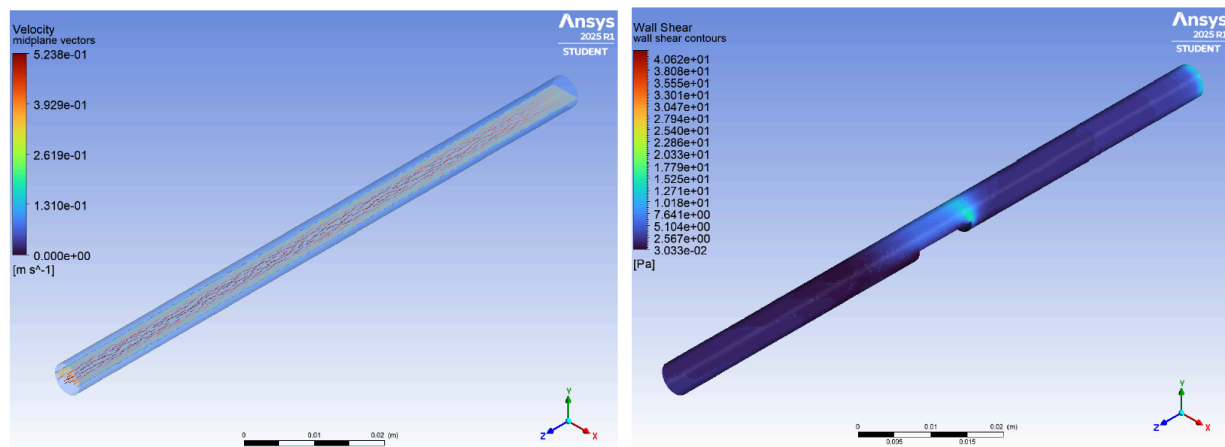


Figure 3: Wall Shear Stress of Normal Healthy Artery vs Artery With an Occlusion Due to Plaque Buildup

The healthy artery's results matched theoretical expectations for laminar pipe flow (Poiseuille profile), providing confidence in the validity of the simulation. The velocity, acceleration, and WSS spike in the plaque model were also expected based on physical principles. These sanity checks confirmed that the simulation reliably reflects physiologic behavior.

This CFD simulation demonstrates how plaque buildup alters hemodynamics in arteries. The increased velocity and wall shear stress at the site of narrowing align with clinical findings that

stenotic regions are high-risk zones for further vascular complications. Elevated WSS may lead to endothelial dysfunction, while regions of flow separation downstream from the plaque could encourage low-shear environments prone to thrombosis or further plaque deposition.